

Neel Shah

Senior Data Engineer

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Professional Summary:

Senior Data Engineer with over 10 years of hands-on experience building and managing scalable data solutions across Azure, AWS. Skilled in developing robust ETL pipelines, optimizing data warehousing, and delivering advanced analytics. My expertise spans across fintech, e-commerce, and social media, leveraging tools like Microsoft Fabric, Microsoft Synapse, Databricks, Snowflake, Spark, and Python to drive insights and improve operations. I thrive in agile, cross-functional teams, aligning data strategies with business goals to support smarter decision-making and deliver measurable impact.

Technical Skills:

Programming Languages: Python, SQL, JavaScript, Scala, PL/SQL, T-SQL

Cloud Platforms: AWS (EC2, S3, EMR, Redshift), GCP (BigQuery, Cloud SQL, Dataflow), Azure (Blob Storage, Synapse Analytics, Data Factory, SQL Database)

Big Data Technologies: Spark, Hadoop (HDFS, MapReduce)

Data Warehousing: Microsoft Fabric, Azure Synapse, Databricks, Snowflake, MSSQL, Big query.

Data Processing & Integration: Apache Airflow, Airbyte, Apache NiFi, Keboola, dbt, Apache Kafka, Pub/Sub

Data Visualization: Looker, Power Bi, Holistics, Tableau

Development & DevOps: Docker, Kubernetes, Terraform, Git, GitHub Actions, GitHub, GitLab, Bitbucket

Security & Compliance: IAM Security, HIPAA Compliance, LDAP

Software Development Practices: Agile, TDD, BDD, CI/CD

Professional Experience:

Stripe, Inc. San Francisco, CA (Remote)

Jan 2023 - Present

Senior Data Engineer

- Designed and implemented a scalable, cloud-native data architecture on AWS and Snowflake using Medallion Architecture, streamlining raw-to-insight data flow to support real-time financial analytics and risk modeling.
- Built end-to-end ELT pipelines using Python, SQL, and dbt to power Stripe's internal data products, ensuring accurate reconciliation of transaction and payment data across services.
- Developed and optimized high-performance ELT workflows in Snowflake and Databricks, applying advanced query tuning to cut financial query latency by over 50%.
- Led the integration of behavioral and transactional datasets from diverse platforms using Kafka, Spark, and Airflow, enabling customer-level analytics, fraud detection, and credit risk evaluation.
- Automated data quality checks and pipeline monitoring with Apache Airflow and Python, improving reliability of critical reporting pipelines used in compliance and finance operations.
- Collaborated with product and risk teams to design star schema-based data models in Snowflake, optimizing access to payment events, customer metadata, and financial transactions for downstream analytics.
- Implemented robust role-based data access policies and column-level security in Snowflake to support Stripe's compliance with SOC2 and PCI-DSS requirements.
- Mentored junior engineers on dbt modeling standards, ELT best practices, and Medallion Architecture, increasing team efficiency and code reliability in production pipelines.

Reddit, Inc. San Francisco, CA

Dec 2021 – Jan 2023

Lead Data Engineer

- Led Reddit's data platform modernization by rebuilding legacy pipelines into scalable, modular ELT frameworks using dbt, Airflow, and Snowflake.
- Designed and managed real-time streaming systems with Kafka and Spark Structured Streaming to support feed ranking, content tagging, and user activity tracking.

- Developed event-driven data models supporting Reddit's ads platform, personalized recommendations, and trust & safety dashboards.
- Spearheaded CI/CD automation for data pipelines, integrating build, quality checks, security scans, and deployment using GitHub Actions and internal tools.
- Advised teams on data governance, including PII handling, column-level access policies, and lineage documentation using tools like Amundsen and Great Expectations.
- Worked closely with data scientists to prototype and deploy machine learning-ready datasets for experimentation and modeling workflows.
- Mentored a team of data engineers, introduced coding standards, code reviews, and performance tuning workshops to improve team productivity.

Reddit, Inc. San Francisco, CA

Jun 2017 – Dec 2021

Senior Data Engineer

- Built and maintained distributed data pipelines on AWS using Spark, EMR, Redshift, and S3, processing billions of user interactions daily.
- Created and maintained star schema-based models in Redshift and Snowflake to support product, ads, and community teams with performant, scalable data access.
- Developed scalable reporting solutions and real-time dashboards for experimentation, A/B testing, and growth analytics using Looker and Redash.
- Collaborated with cross-functional teams to instrument product events, normalize behavioral tracking, and integrate them into the core data models.
- Automated data validation, testing, and monitoring across pipelines to meet data SLAs and reduce pipeline failures by 40%.
- Acted as a liaison between data engineering and product analytics, aligning data modeling practices with rapid experimentation needs.
- Piloted the use of dbt and modular SQL frameworks, laying the groundwork for future data model scalability across the company.

Reddit, Inc. San Francisco, CA

Oct 2015 – Jun 2017

Data Engineer

- Built Reddit's foundational data infrastructure on AWS, configuring secure VPCs, EC2, RDS, and S3 for scalable batch data processing.
- Created ingestion pipelines to centralize Reddit post, comment, vote, and session data into the company's first data lake.
- Developed internal tools with Flask and Python for data exploration, empowering teams to query and visualize key product metrics.
- Implemented alerting and monitoring for ETL failures and latency using CloudWatch and PagerDuty, improving pipeline reliability.
- Supported the data needs of teams across growth, safety, and platform engineering with custom SQL models and reporting layers.
- Helped define internal policies for data retention, privacy, and user anonymization, aligning with GDPR and internal compliance efforts.
- Delivered training sessions on SQL, AWS, and data warehouse best practices to analysts and product managers.

Education:

Bachelor of Science in **Computer Science**,
University of Texas at Austin, Austin, TX

2011 - 2015